

RAPHAEL PENZ

Gameplay Programmer | Software Developer

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PROFILE

Gameplay programmer and software developer with a B.Sc. in Computer Science (Games Engineering) from the Technical University of Munich, focused on gameplay systems, software architecture and Python/C++ development. Solo developer of a published RPG/visual-novel hybrid with 2,500+ downloads and a substantial interconnected gameplay codebase. Additional experience in performance-oriented C++ simulation software, SIMD optimization and scalable benchmark infrastructure.

TECHNICAL PROFILE

Programming Languages	Python • C++ • C# • Java
Game Development	Ren'Py • Unity • Unreal Engine • Gameplay systems • Combat, quest, inventory, character and progression systems
Architecture	Modular gameplay architecture • Data-driven systems • Save/update compatibility
Performance	SIMD optimization • AoS/SoA memory layouts • Scalable simulations
Tools	Git • Visual Studio • CMake • Linux
Languages	German - Native • English - C2

PROJECT EXPERIENCE

Harmony of the Stars - Solo Developer

September 2024 - present

RPG/visual novel hybrid | Python, Ren'Py | published solo project | Portfolio & code samples

- Developed and publicly released a solo-developed RPG/visual-novel hybrid with 2,500+ downloads.
- Designed and implemented interconnected gameplay systems including turn-based combat, inventory/equipment, quests, progression, relationships, character scheduling and world navigation.
- Built a modular, data-driven gameplay architecture to support system interaction, long-term maintainability and continued expansion.
- Designed a save- and update-compatible project structure so new content and system changes can continue to support existing save files.
- Maintained 23,000+ lines of gameplay/application code within a total project codebase of 47,000+ lines.
- Full ownership of programming, systems design, debugging, software architecture, release management and technical project organization.

PROJECT EXPERIENCE - C++ / PERFORMANCE

AutoPas - Bachelor's Thesis, Technical University of Munich

October 2021

C++ particle simulation framework | Discrete Element Method | SIMD | AoS/SoA

- Implemented a new particle type based on the Discrete Element Method (DEM) in the existing C++ AutoPas framework.
- Developed pairwise force-calculation functors for AoS and SoA memory layouts, including SIMD optimization.
- Extended the simulation infrastructure and integrated DEM simulations into the existing molecular-dynamics benchmark suite.
- Evaluated scalability and runtime performance using simulations with up to 1,000,000 particles.
- Compared physical behavior and runtime characteristics against existing molecular-dynamics implementations.

EDUCATION

B.Sc. Computer Science: Games Engineering - TUM

Oct 2017 - Sep 2024

Technical University of Munich, Munich

M.Sc. Computer Science: Games Engineering - TUM

Oct 2024 - Sep 2025

Technical University of Munich, Munich | 87/120 ECTS completed

PROFESSIONAL EXPERIENCE

Luxury Boutique Hotel Himmlhof - Hotel Employee (part-time)

August 2021 - present

St. Anton am Arlberg, Tyrol

- Part-time role in daily hotel operations with additional IT support, including PC setup, Windows/software troubleshooting, printer configuration and website maintenance.
- Supported guest service, reception and organizational tasks while communicating reliably with guests and colleagues in a seasonal business environment.
- Adapted flexibly across operational and technical tasks according to day-to-day requirements.

Austrian Red Cross - Civilian Service / Paramedic

September 2016 - June 2017

Landeck, Tyrol

- Worked as a paramedic and supported the dispatch center with call handling, organization of patient transports and coordination of drivers and vehicles.

PORTFOLIO & CODE SAMPLES

Portfolio: theh0bbit.github.io • Code samples: github.com/TheH0bbit/harmony-of-the-stars-code-samples